seekdb Hybrid Retrieval Field Notes

Author: ResearchFlow demo corpus. Scope: seekdb v1.4.0.0 on OceanBase 4.3.5.3.

Page 1 records the baseline environment used for every measurement in this document. The server reports VERSION() as 5.7.25-OceanBase seekdb-v1.4.0.0 and listens on port 2881. The database used for all examples is named test.

第 1 页（中文摘要）：基线环境为 seekdb v1.4.0.0，底层 OceanBase 4.3.5.3，监听端口 2881，示例数据库名为 test。SELECT VERSION() 的返回值是 5.7.25-OceanBase seekdb-v1.4.0.0。

Page 2 - Vector index configuration

seekdb stores dense vectors in a VECTOR(n) column. The demo corpus uses n = 384, matching the all-MiniLM-L6-v2 ONNX embedding shipped inside the official pyseekdb SDK.

The index type is HNSW and the distance metric is cosine. The DDL fragment generated by pyseekdb is: WITH (DISTANCE=cosine, TYPE=hnsw, LIB=vsag).

Approximate nearest neighbour search uses this ordering clause: ORDER BY l2_distance(embedding, :query_vector) APPROXIMATE LIMIT 10 PARAMETERS(ef_search=64).

Measured cosine distance for the query 'hybrid retrieval vector and full text' against the best matching row was 0.3753, giving a similarity of 0.6247.

第 2 页（中文摘要）：向量列类型为 VECTOR(384)，索引类型 HNSW，距离度量 cosine。
查询 hybrid retrieval vector and full text 的最佳匹配余弦距离是 0.3753，
对应相似度 0.6247。

Page 3 - Full text analyzers

seekdb supports several FULLTEXT analyzers. The three verified in this project are ik, ngram and space. The DDL is:
CREATE FULLTEXT INDEX ix_name ON tbl (col) WITH PARSER ik

For the Chinese needle '混合召回' the ik analyzer ranked the row that actually contains that phrase first. The space analyzer ranked a different row first because it does not segment Chinese text at all.

Conclusion recorded on page 3: use the ik analyzer as the default for a bilingual Chinese and English research corpus, and keep the analyzer name configurable so ngram can be substituted without a code change.

第 3 页（中文摘要）：全文分析器共有 ik、ngram、space 三种经过验证。中文查询「混合召回」下 ik 分析器把真正包含该短语的记录排在第一位，space 分析器因为不切分中文而排错。结论：双语科研语料默认使用 ik 分析器。

Page 5 - Reciprocal Rank Fusion

Hybrid retrieval in seekdb fuses a full text leg and a vector leg with Reciprocal Rank Fusion. The score of a document is the sum over both legs of 1 / (rank_constant + rank), with rank_constant = 60 by default.

A row placed first in both legs scores 1/61 + 1/61 = 0.0328. A row placed first in one leg and second in the other scores 1/61 + 1/62 = 0.0325. A row present in only one leg at rank 1 scores 1/61 = 0.0164.

Those three values - 0.0328, 0.0325 and 0.0164 - are exactly the distances returned by collection.hybrid_search with rank={'rrf': {}} on the probe corpus, which is how the RRF implementation was confirmed end to end.

第 5 页（中文摘要）：混合检索用互惠排名融合 RRF 合并全文腿与向量腿，公式为 1 / (rank_constant + rank) 求和，默认 rank_constant = 60。
一条记录如果在两条腿里都排第一，融合分数是 1/61 + 1/61 = 0.0328；一条腿第一、另一条腿第二是 0.0325；只在一条腿里排第一是 0.0164。